



Leafline

DETECTING TREE CROWNS FROM AERIAL IMAGES

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INTRODUCTION

How well can an existing model perform in a real-world application?

Fine-tune a segmentation model to reliably detect individual tree crowns in Kiel city

Instance Segmentation – detect presence and number of tree crowns



LITERATURE REVIEW

Freudenberg et al., 2022

Tree Crown Segmentation with U-Net architecture. Basis for our model.

Teng et al., 2025

Shows that introducing height data improves tree crown segmentation by adding structural information to spectral images.

Khan et al., 2025

Python package for tree crown segmentation. Includes pretrained weights from Freudenberg et al. 2022, which serves as our baseline model.

Rottensteiner et al., 2022

Systematic analysis of methods for urban object extraction. Standard evaluation metrics.

DATASET OVERVIEW



9 RGBI orthophotos of variable size
(mean area 83.000 m²)



3.178 manually annotated trees



Seasonal variability – images from
spring and from summer



Variable image resolution – 20cm vs.
7.5cm



DATASET OVERVIEW

Training



4 images



73% area



2229 trees

Validation



3 images



5% area



217 trees

Test



2 images



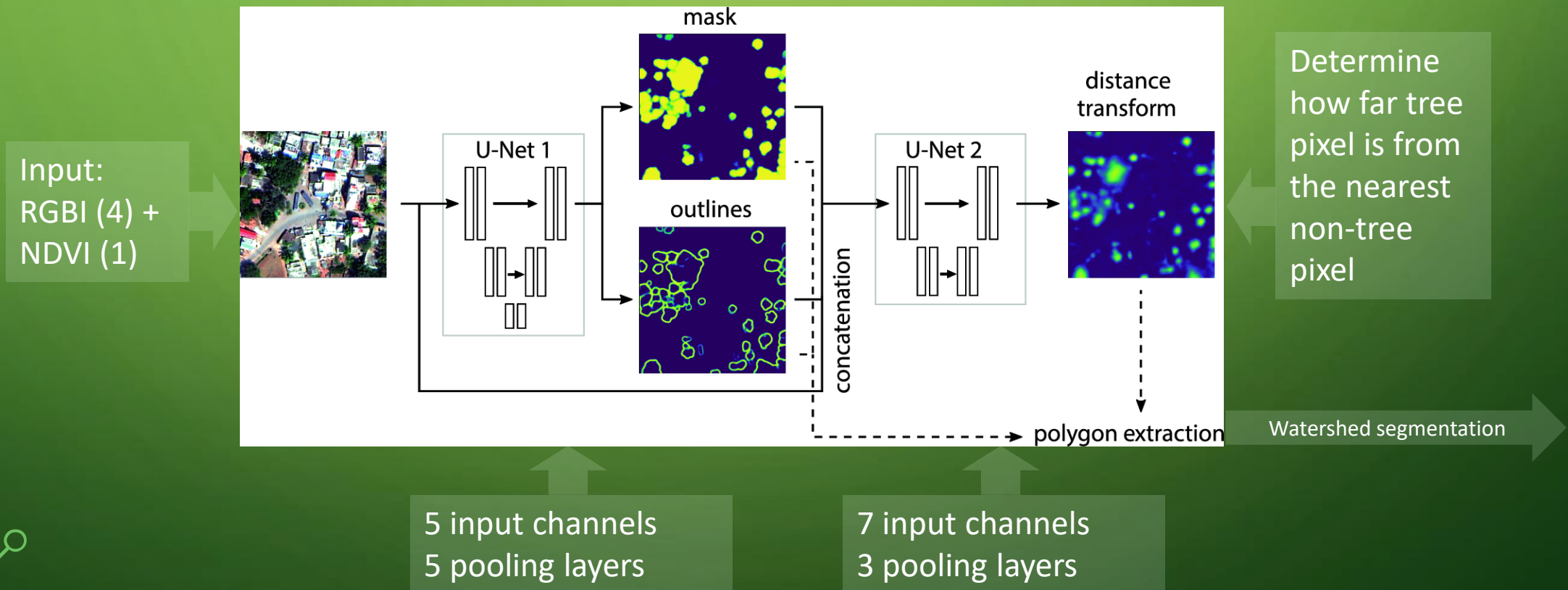
22% area



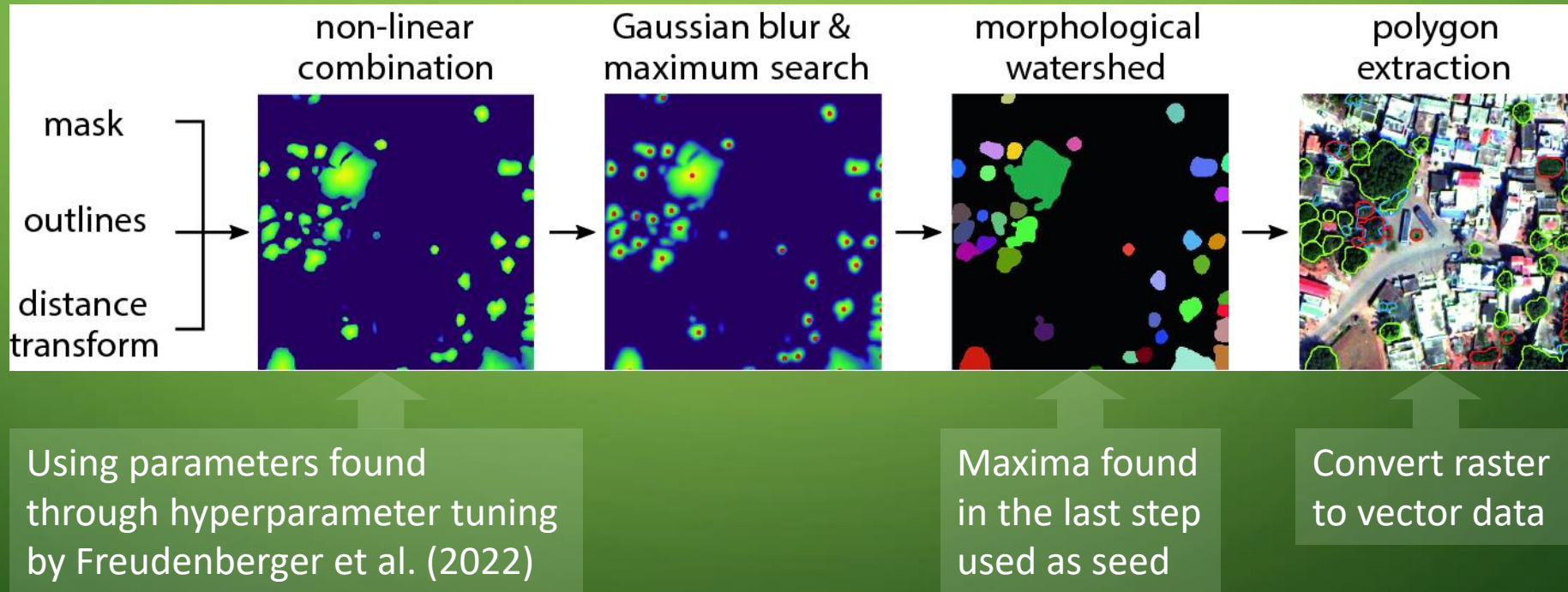
723 trees

BASELINE MODEL

U-Net with a ResNet 18 Backbone
pretrained on ImageNet and datasets from Freudenberg et al. 2022



BASELINE MODEL



PIPELINE

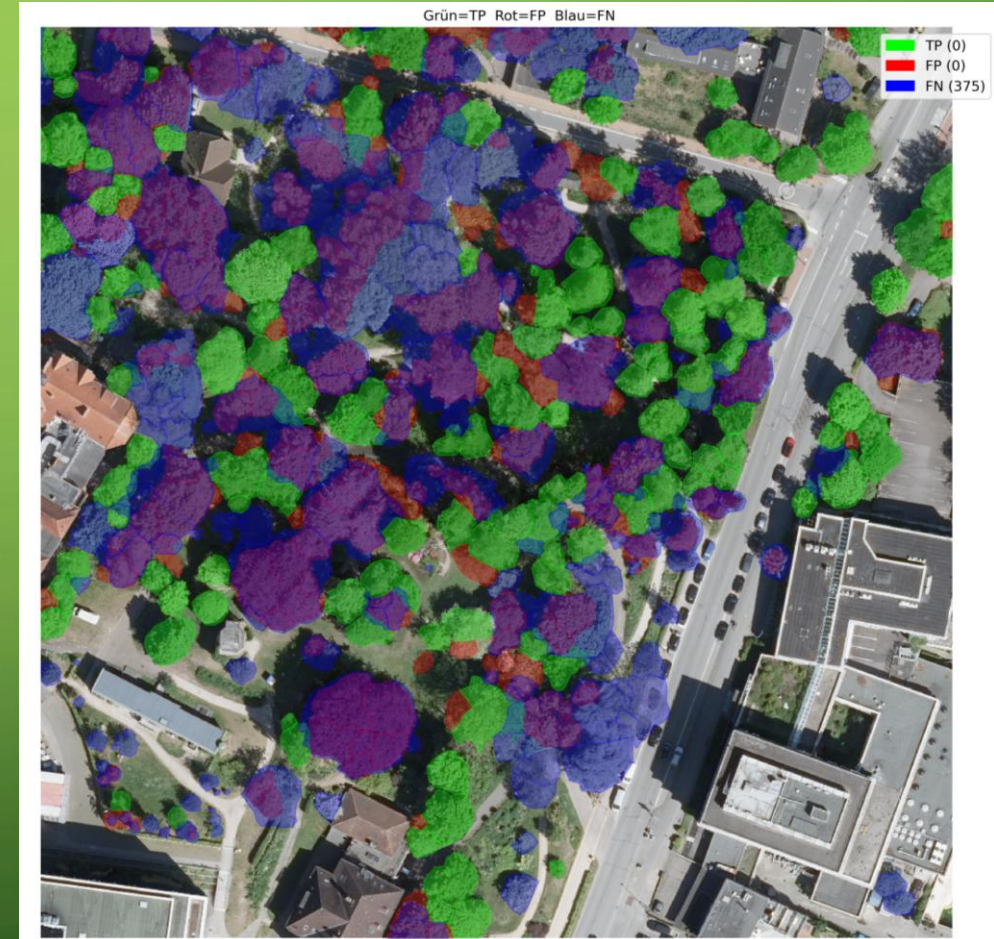
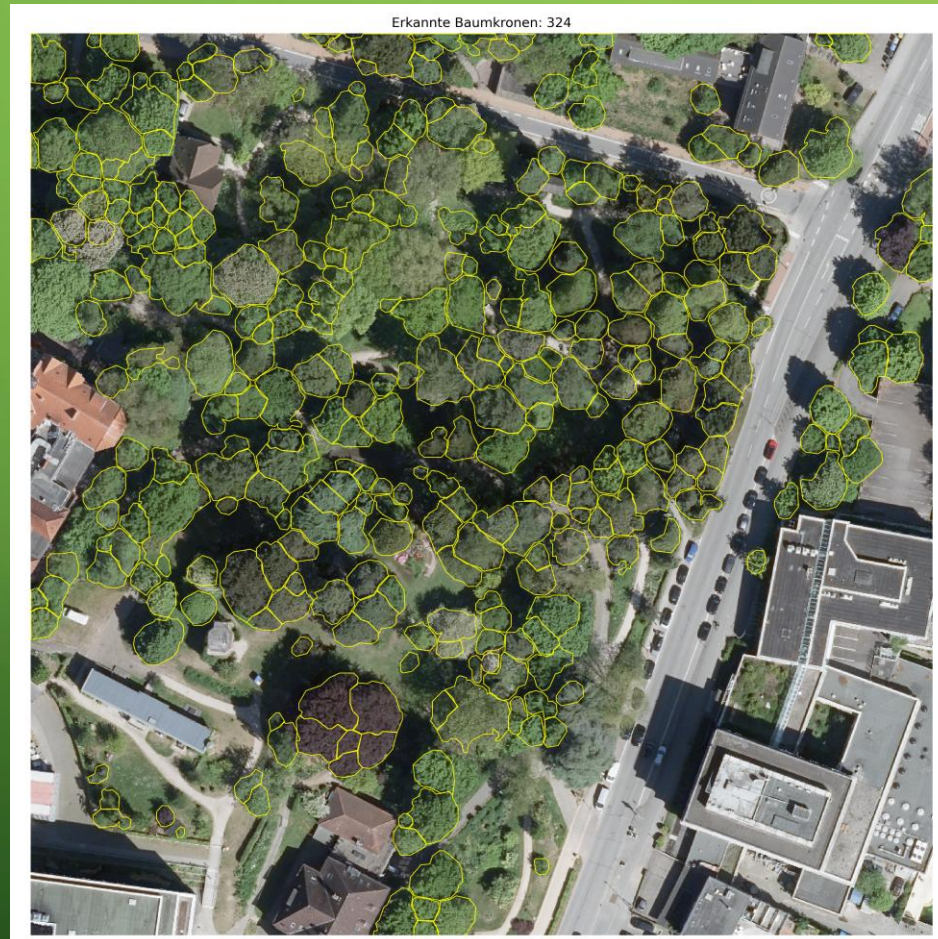
- 
- Load images
 - Calculate and add NDVI
 - Create overlapping patches (size: 256x256, stride: 128)
 - Normalize RGBI [0,1], (NDVI already normalized)
 - DataLoader
 - Postprocessing and Evaluation

MODEL EVALUATION

- Instance Evaluation (IoU Matching)
 - Find best fitting GT polygon for prediction (greedy, sorted by IoU)
 - If best IoU $\geq$ IoU threshold TP, otherwise FP
 - Non-matched GT polygons are FN

Precision	$TP / (TP + FP)$	how many found trees are real?
Recall	$TP / (TP + FN)$	how many real trees were found?
F1	$2 * Precision * Recall / (Precision + Recall)$	

BASELINE RESULTS



BASELINE RESULTS

Area	Resolution	Season	Prediction	Ground Truth	True Positive	False Positive	False Negative	Precision	Recall	F1
BotGarten	20cm	summer	324	357	129	195	228	0.398	0.361	0.379
BotGarten	20cm	spring	83	357	18	65	339	0.217	0.050	0.082
BotGarten	7.5cm	spring	68	357	13	55	344	0.191	0.036	0.061
HoernNord	20cm	summer	139	375	74	65	301	0.532	0.197	0.288
HoernNord	20cm	spring	0	375	0	0	375	0.000	0.000	0.000
HoernNord	7.5cm	spring	0	375	0	0	375	0.000	0.000	0.000

- Seasonality has a bigger impact on model performance than image resolution

FINETUNING

- Training with patience stopping. Saving best checkpoint
- Loss Function: $\text{loss_mask} + \text{loss_outline} + 2 * \text{loss_distance}$
- Augmentations:
 - Hflip, vflip (each 50%)
 - Spectral gaussian blur on channel 0-4 (sigma = 0.02)
 - RGBI: brightness *0.75-1.25, contrast shift +/-0.1
 - NIR: scaling *0.1-1.0
 - NDVI: scaling *0.2-1.0

in_channels	6
lr	1.0e-4
batch_size	8
num_workers	4
max_epochs	50
patience	10

FINETUNING RESULTS

Area	Resolution	Season	Prediction	Ground Truth	True Positive	False Positive	False Negative	Precision	Recall	F1
BotGarten	20cm	summer	179	357	88	91	269	0.492	0.246	0.328
BotGarten	20cm	spring	52	357	8	44	349	0.154	0.022	0.039
BotGarten	7.5cm	spring	508	357	75	433	282	0.148	0.210	0.173
HoernNord	20cm	summer	130	375	69	61	306	0.531	0.184	0.273
HoernNord	20cm	spring	25	375	4	21	371	0.160	0.011	0.020
HoernNord	7.5cm	spring	218	375	37	181	338	0.170	0.099	0.125

FINETUNING RESULTS

Finetuning including all datasets (summer + spring + nDOM)

Summer (20cm)

↓ Precision

↓ Recall

↓ F1

➤ Overall slightly worse

Spring (20cm)

↘ Precision

↘ Recall

↘ F1

➤ BotGarten got much worse, Hörn slightly better

Spring (7.5cm)

↑ Precision

↑ Recall

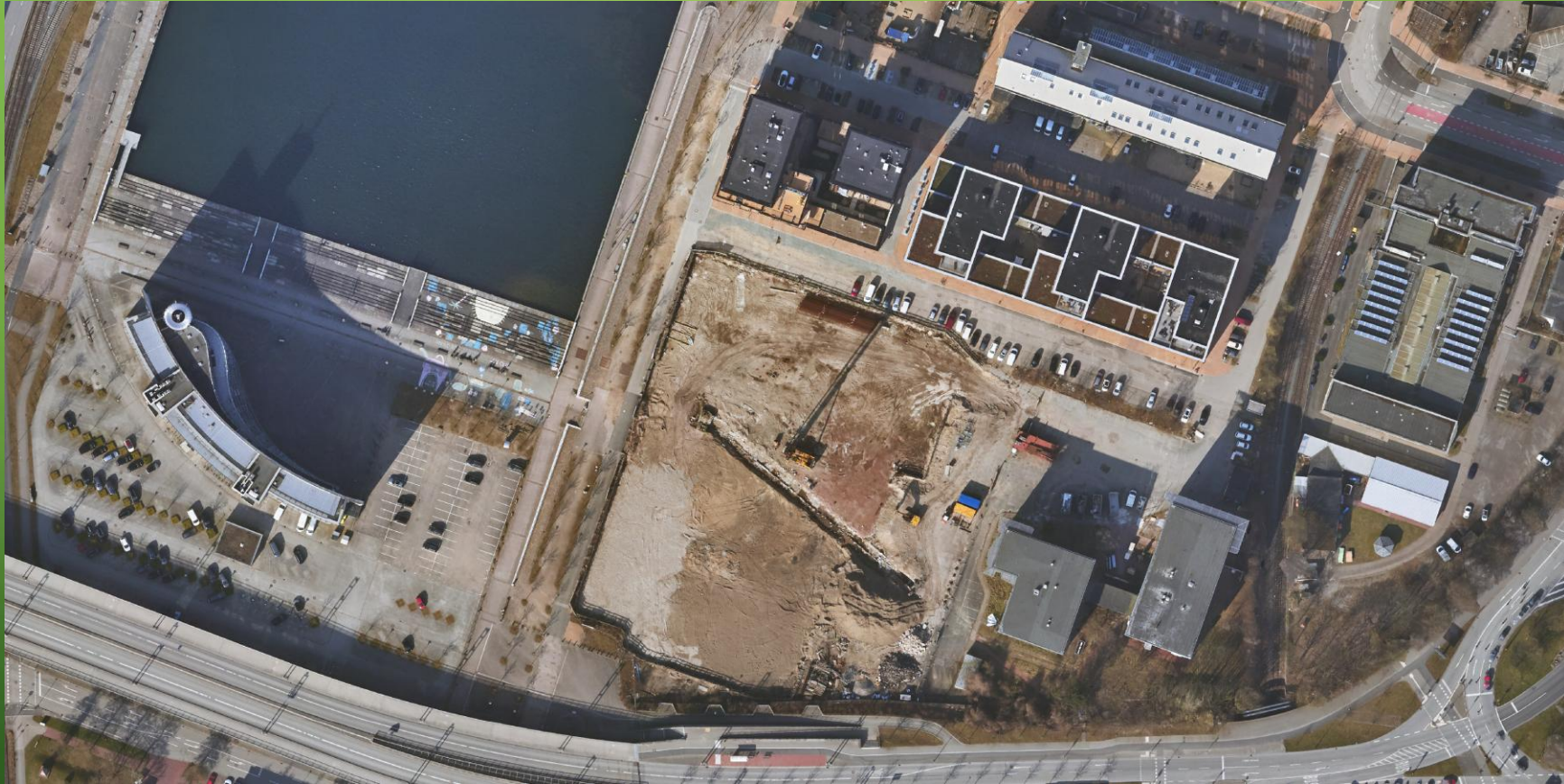
↑ F1

➤ Model tends to over-segment trees

DISCUSSION



DISCUSSION



DISCUSSION



MAIN CHALLENGES

- Small and understory trees are very difficult to predict with 2.5D data
- Data preparation is very time intensive, especially when working with restricted data
- Working with an existing model architecture can be surprisingly challenging
- Changing multiple variables needs lots of tests to understand effects

SUMMARY AND TAKEAWAY

- Baseline performs much better in summer than spring (expected)
- Finetuning with all data leads the model to oversegment trees (why?)
- Finetuning including spring data improves recognition in spring images
- Need to separate variables to find out, what exactly influences the changes
- Need to optimize hyperparameters (especially watershed segmentation)

NEXT STEPS

Separate finetuning variables to better understand effects

- Better to include both seasons/resolutions or make separate models?
- Test inclusion of nDOM as a separate variable to isolate influence

Hyperparameter tuning

- Increase percentage of validation data for better hyperparameter tuning
- Tune hyperparameters for both U-Net and watershed segmentation



THANK YOU FOR YOUR ATTENTION